آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 7

Functions of Two Random Variables

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Learning Objectives

After completing this module, students will be able to:

1. Derive the distribution of $Z = X + Y$ using convolution
2. Find distributions of differences, products, ratios, max, and min
3. Apply the bivariate Jacobian transformation method
4. Connect these results to engineering applications (signal+noise, reliability)

7.1 Introduction

Engineering systems combine random variables:

- **Received signal** = transmitted + noise: $R = S + N$
- **Total interference** = sum of multiple sources: $I = I_1 + I_2 + \ldots + I_n$
- **System lifetime** = minimum of component lifetimes: $T = \min(T_1, T_2)$
- **SNR** $= \dfrac{\text{signal power}}{\text{noise power}}$: ratio of RVs

We need methods to find distributions of **functions of two (or more) RVs**.

7.2 Sum of Two Random Variables: $Z = X + Y$

CDF Method

$F_Z(z) = P(X + Y \le z) = \int\int_{x+y\le z} f_{X,Y}(x,\, y)\, dx\, dy$

Convolution Formula (Independent Case)

If X and Y are **independent**:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z - x)\, dx = (f_X * f_Y)(z)$$

This is the **convolution** of f_X and f_Y.

Derivation

$F_Z(z) = P(X + Y \le z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(y)\, dy\, dx$
$= \int_{-\infty}^{\infty} f_X(x) F_Y(z - x)\, dx$

Differentiating: $f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx$ ✔

Important Special Cases

Sum of two Gaussians (independent):
$X \sim N(\mu_1,\, \sigma_1^2),\, Y \sim N(\mu_2,\, \sigma_2^2) \to Z = X + Y \sim N(\mu_1 + \mu_2,\, \sigma_1^2 + \sigma_2^2)$

Sum of two exponentials (same rate):
$X,\, Y \sim \text{Exp}(\lambda)$ independent $\to Z \sim \text{Gamma}\left(2,\, \frac{1}{\lambda}\right) = \text{Erlang}(2,\, \lambda)$

Sum of two Poissons:
$X \sim \text{Poisson}(\lambda_1),\, Y \sim \text{Poisson}(\lambda_2)$ independent $\to Z \sim \text{Poisson}(\lambda_1 + \lambda_2)$

Sum of two uniforms:
$X,\, Y \sim \text{Uniform}(0,\, 1)$ independent $\to Z$ has triangular distribution on [0,2]

Engineering Example: Signal + Noise

Transmitted signal $S \sim N(A,\, \sigma_s^2)$, noise $N \sim N(0,\, \sigma_n^2)$, independent.
Received: $R = S + N \sim N(A,\, \sigma_s^2 + \sigma_n^2)$
The received signal is still Gaussian — with increased variance (reduced SNR).

7.3 Difference: $Z = X - Y$

For independent X, Y:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(x - z)\, dx$$

Gaussian case: $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim N(\mu_2, \sigma_2^2) \to Z = X - Y \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$

Note: variances still ADD (not subtract) because $\text{Var}(-Y) = \text{Var}(Y)$.

Engineering Example: Differential Measurement

Two sensors measure same quantity: $X_1 = \theta + N_1$, $X_2 = \theta + N_2$.
Difference: $X_1 - X_2 = N_1 - N_2 \sim N(0, \sigma_1^2 + \sigma_2^2)$
The difference eliminates the common signal but doubles the noise variance.

7.4 Product: $Z = XY$

For independent X, Y:

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{|x|} f_X(x) \cdot f_Y(z/x)\, dx$$

Engineering Example: Power

Voltage V and current I in a linear circuit: $P = V \cdot I$.
If V and I have known joint distribution, we can find the power distribution.

7.5 Ratio: $Z = \frac{X}{Y}$

For independent X, Y:

$$f_Z(z) = \int_{-\infty}^{\infty} |y| \cdot f_X(zy) \cdot f_Y(y)\, dy$$

Engineering Example: SNR

$$\text{SNR} = \frac{\text{Signal Power}}{\text{Noise Power}} = \frac{P_s}{P_n}$$

If both are chi-squared distributed (sum of squared Gaussians), the ratio follows an **F-distribution**.

7.6 Maximum: $Z = \max(X, Y)$

CDF Approach

$F_Z(z) = P(\max(X, Y) \leq z) = P(X \leq z \text{ AND } Y \leq z)$

If independent: $F_Z(z) = F_X(z) \cdot F_Y(z)$

PDF: $f_Z(z) = f_X(z)F_Y(z) + F_X(z)f_Y(z)$

Engineering Example: Parallel Redundancy

System works if AT LEAST ONE component works. System fails only when ALL fail.
System lifetime $= \max(T_1, T_2)$ for parallel redundant components.
For $T_1, T_2 \sim \text{Exp}(\lambda)$ independent:
$F_Z(z) = (1 - e^{-\lambda z})^2$ for $z \geq 0$
$f_Z(z) = 2\lambda e^{-\lambda z}(1 - e^{-\lambda z})$
Mean lifetime: $E[\max] = \frac{3}{2\lambda} > \frac{1}{\lambda} = E[\text{single component}]$ (reliability improved!)

7.7 Minimum: $Z = \min(X, Y)$

Survival Function Approach

$P(Z > z) = P(\min(X, Y) > z) = P(X > z \text{ AND } Y > z)$

If independent: $P(Z > z) = P(X > z) \cdot P(Y > z) = [1 - F_X(z)][1 - F_Y(z)]$

CDF: $F_Z(z) = 1 - [1 - F_X(z)][1 - F_Y(z)]$

Engineering Example: Series System

System fails when FIRST component fails. System lifetime $= \min(T_1, T_2)$.
For $T_1 \sim \text{Exp}(\lambda_1)$, $T_2 \sim \text{Exp}(\lambda_2)$ independent:
$P(Z > z) = e^{-\lambda_1 z} \cdot e^{-\lambda_2 z} = e^{-(\lambda_1+\lambda_2)z}$
Therefore: $\min(T_1, T_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$

Key result: Failure rates ADD in series systems. Mean lifetime $= \frac{1}{\lambda_1+\lambda_2} < \min\left(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}\right)$.

7.8 General Bivariate Transformation (Jacobian Method)

Setup

Given $(X,\, Y)$ with known joint PDF, find the joint PDF of $(U,\, V)$ where:

- $U = g_1(X,\, Y)$
- $V = g_2(X,\, Y)$

Procedure

1. Solve for the inverse: $X = h_1(U,\, V)$, $Y = h_2(U,\, V)$
2. Compute the Jacobian determinant:

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

3. Apply:
$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v)) \cdot |J|$$
4. If only U is needed, marginalize out V.

Example: Sum and Difference

$$U = X + Y,\, V = X - Y \rightarrow X = \frac{U+V}{2},\, Y = \frac{U-V}{2}$$
$$J = \left|\frac{\partial x}{\partial u} \cdot \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \cdot \frac{\partial y}{\partial u}\right| = \left|\frac{1}{2}\left(-\frac{1}{2}\right) - \frac{1}{2}\left(\frac{1}{2}\right)\right| = \frac{1}{2}$$
$$f_{U,V}(u,\, v) = \frac{1}{2} \cdot f_{X,Y}\left(\frac{u+v}{2},\, \frac{u-v}{2}\right)$$

7.9 Practice Problems

Problem 1

$X \sim \text{Exp}(1)$, $Y \sim \text{Exp}(1)$, independent. Find the PDF of $Z = X + Y$.

Solution Convolution: $f_Z(z) = \int_0^z e^{-x} \cdot e^{-(z-x)}\, dx = \int_0^z e^{-z}\, dx = ze^{-z}$, $z \geq 0$.

This is Gamma$(2,\, 1)$ = Erlang$(2,\, 1)$. $E[Z] = 2$, $\text{Var}(Z) = 2$.

Problem 2

Three components in series with failure rates $\lambda_1 = 0.001$, $\lambda_2 = 0.002$, $\lambda_3 = 0.003$ per hour.

(a) Find system MTTF. (b) Find $P(\text{system survives 100 hours})$.

Solution

(a) MTTF $= \frac{1}{0.001+0.002+0.003} = \frac{1}{0.006} = 166.7$ hours

(b) $P(T_{\text{sys}} > 100) = e^{-0.006\times 100} = e^{-0.6} = 0.549$

Problem 3

$X \sim N(5,\, 4)$, $Y \sim N(3,\, 9)$, independent. Find distribution and $P(X + Y > 12)$.

Solution $Z = X + Y \sim N(8,\, 13)$. $P(Z > 12) = P\left(Z' > \frac{12-8}{\sqrt{13}}\right) = Q(1.109) = 0.1337$.

Next Module: Module 8 — Conditional Expectation and Variance